JAVA COLLECTIONS CASE STUDY 4

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Unique Courses - HashSet

Topic: Set Operations & Course Registration Workflow
Environment: Java SE 17+ / 24

1. Problem Statement & Scenario Description

Scenario: A student can register for multiple courses, but the same course cannot be registered twice. The system supports course enrollment, enrollment status queries, dropping courses, and listing all active enrollments.

#	Required Operation / Method	Implementation Details & Collections Logic
1	**registerCourse(Set courses, String course)**	Inserts course name into the set; ignores repeated registration attempts.
2	**isRegistered(Set courses, String course)**	Checks whether the given course string is currently present in the set.
3	**dropCourse(Set courses, String course)**	Removes the course from the set using courses.remove(course).
4	**displayCourses(Set courses)**	Iterates and prints each enrolled course name on standard output.

2. Collections Framework Concepts & Technical Analysis

- **String Hashing in Collections:** String class provides an optimized hashCode() and equals() implementation for reliable hash set operations.
- **Idempotent Set Mutation:** Adding an existing course has no side effects, and dropping an absent course safely returns false without errors.
- **Unordered Iteration:** HashSet does not guarantee any specific iteration order of courses as entries are placed in buckets by hash.

3. Complete Java Solution Code (Human Written - No Comments)

Clean, idiomatic Java implementation completed according to instructions with zero comments.

Source File: src/collections_case_studies/CourseRegistration.java

```
package collections_case_studies;

import java.util.HashSet;
import java.util.Set;

public class CourseRegistration {

    public static void registerCourse(Set<String> courses, String course) {
        courses.add(course);
    }

    public static boolean isRegistered(Set<String> courses, String course) {
        return courses.contains(course);
    }

    public static void dropCourse(Set<String> courses, String course) {
        courses.remove(course);
    }

    public static void displayCourses(Set<String> courses) {
        for (String course : courses) {
            System.out.println(course);
        }
    }

    public static void main(String[] args) {
        Set<String> courses = new HashSet<>();

        registerCourse(courses, "Java");
        registerCourse(courses, "Python");
        registerCourse(courses, "DBMS");
        registerCourse(courses, "Java");

        System.out.println("Courses:");
        displayCourses(courses);

        System.out.println("Python registered: " + isRegistered(courses, "Python"));

        dropCourse(courses, "DBMS");

        System.out.println("After dropping DBMS:");
        displayCourses(courses);
    }
}
```

4. Program Execution & Verification Output

Console output produced by running the completed driver code matching expected laboratory output:

Console Standard Output (stdout): java collections_case_studies.CourseRegistration

```
Courses:
Java
DBMS
Python
Python registered: true
After dropping DBMS:
Java
Python
```